



CSF3133

WEB-BASED INTERFACE DESIGN

LAB MODULE 6

JAVASCRIPT

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DATE:

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6.1 Introduction to Javascript

Javascript is a versatile programming language widely used to create dynamics and interactive content on websites. Alongside HTML and CSS, it forms the core technologies of web development, enabling developers to build engaging and responsive user experience.

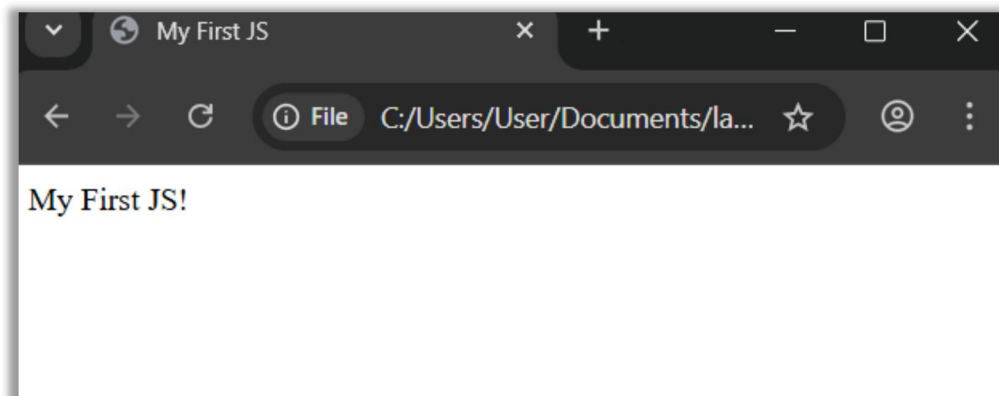
6.2 Objectives

By the end of this lab, students will be able to:

1. Develop familiarity with the Javascript language
2. Learn to use best-practice idions and patterns
3. Understand concepts commonly used in dynamics language programming

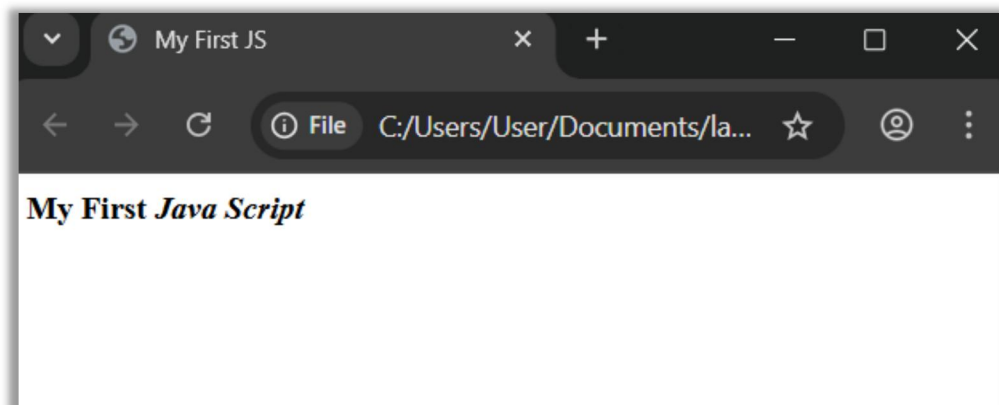
6.3 Basic Javascript

Output:



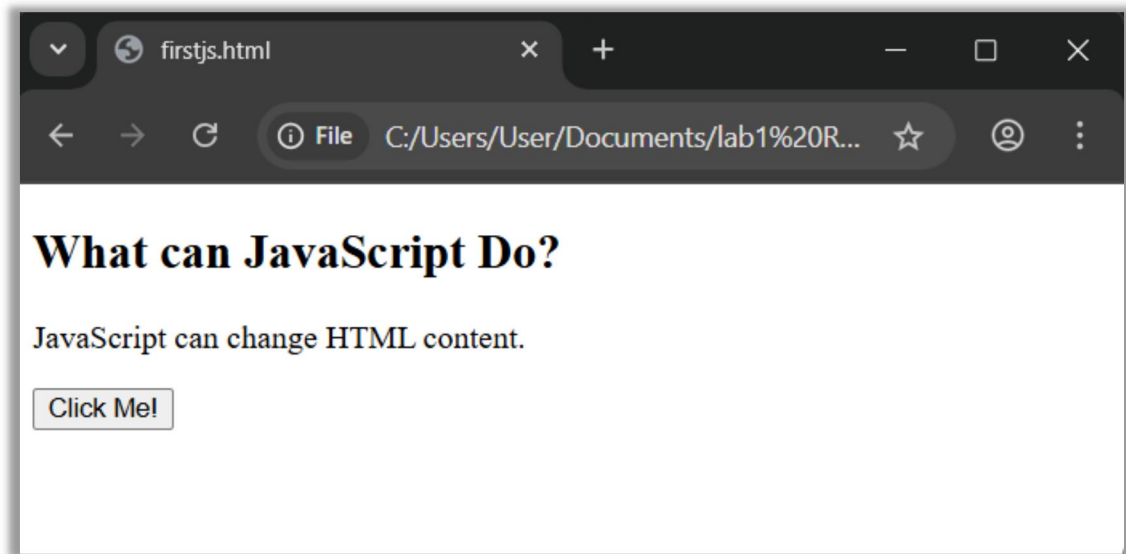
After change the statement `document.write("My First JF!")` into: `document.write("<b> My First </b>" + "<i>Java Script</i>");`

Output:



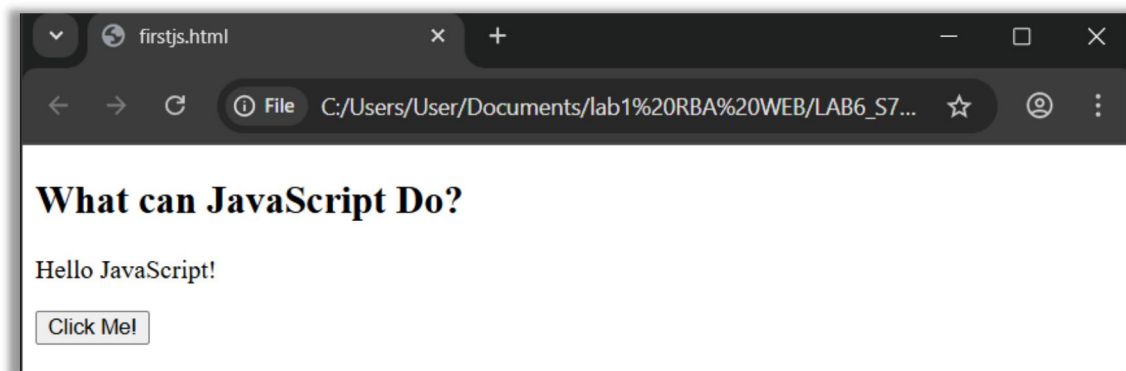
6.4 getElementById()

Output:



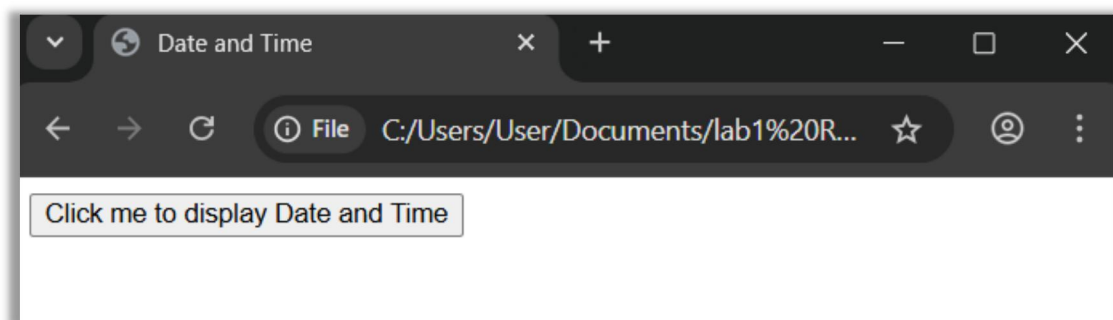
After click the Click Me! Button:

Output:



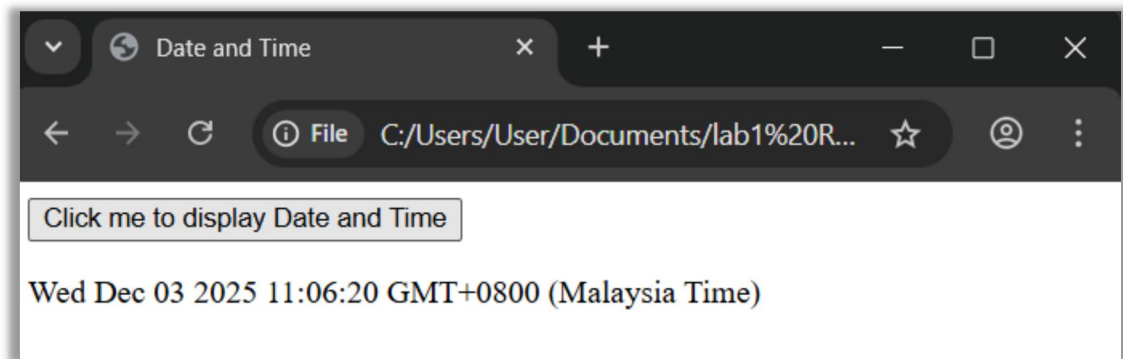
6.5 Date and Time

Output:



After clicking the “Click me to display Date and Time” button:

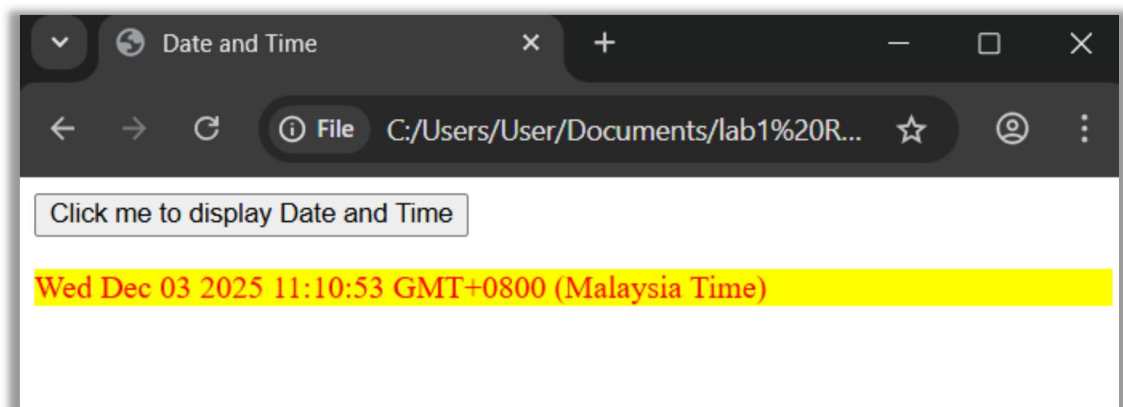
Output:



After try to add these script to change the style of variable (demo):

```
document.getElementById("demo").style.color = "red";  
document.getElementById("demo").style.backgroundColor = "yellow";
```

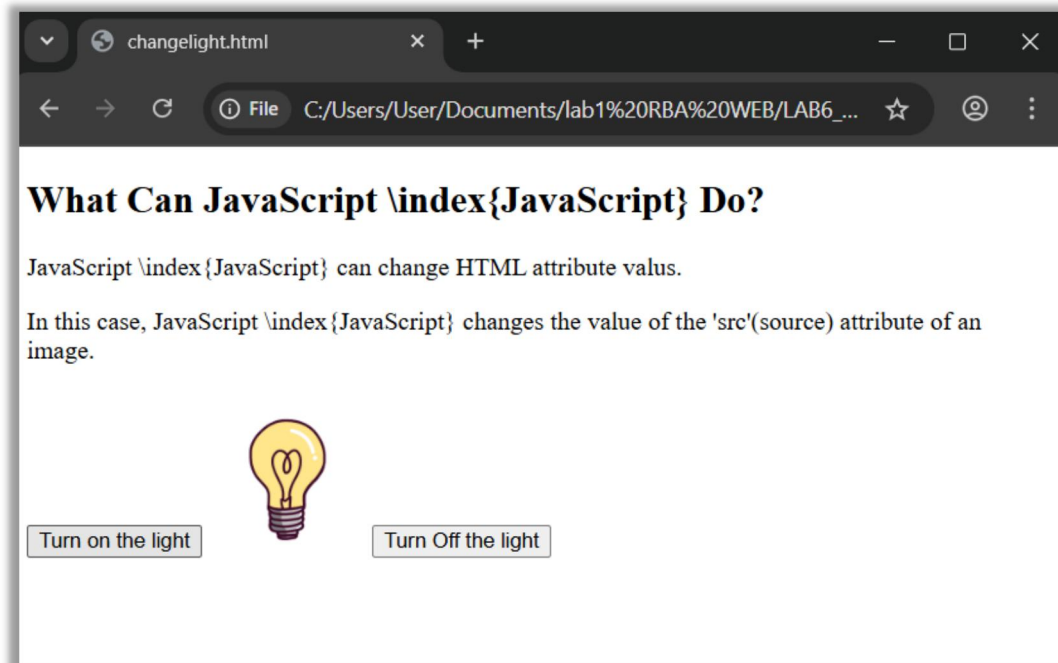
Output:



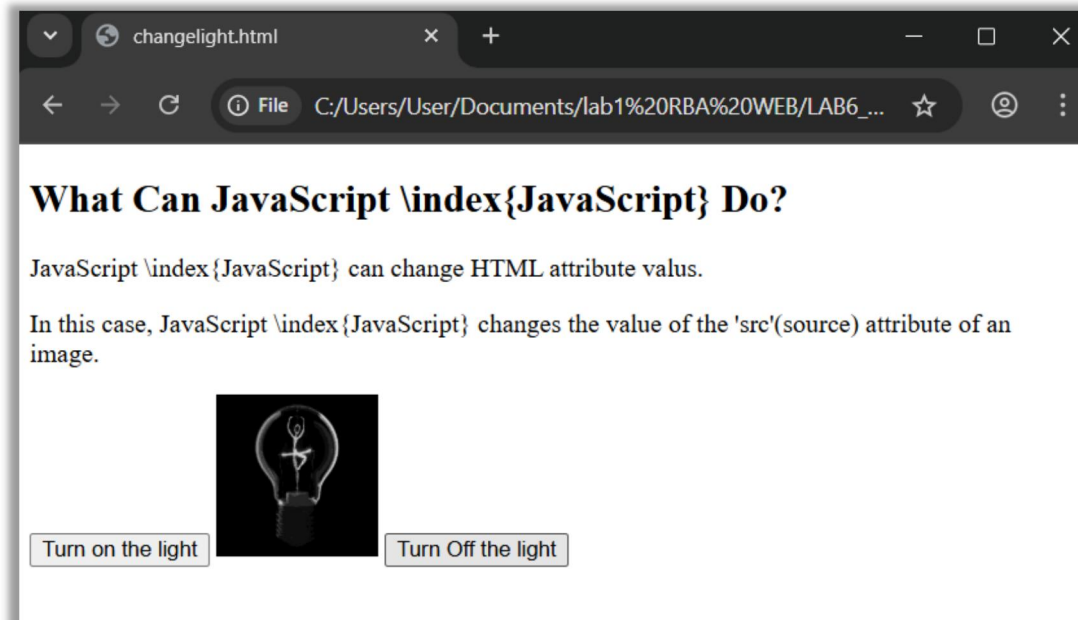
6.6 Change HTML Attribute Values

Output:

When push the “Turn on the light” button:



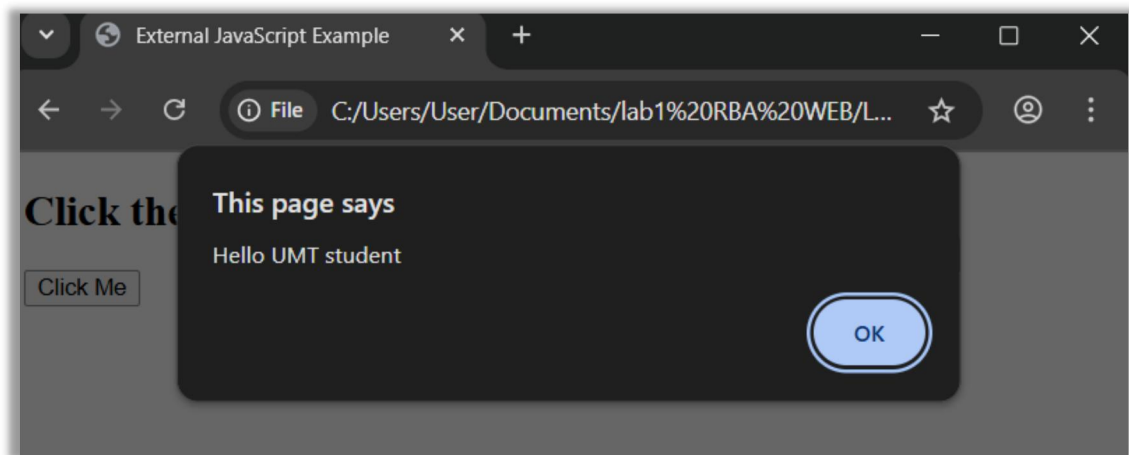
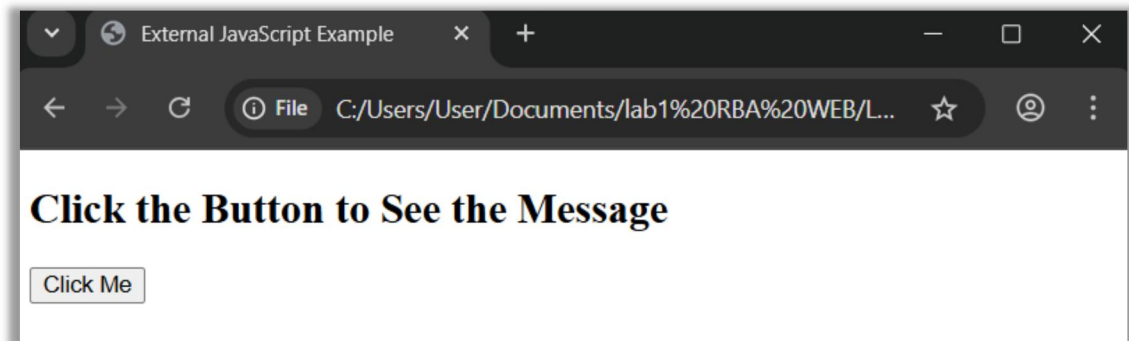
When push the “Turn off the light” button:



6.7 Link to file Javascript (*.js)

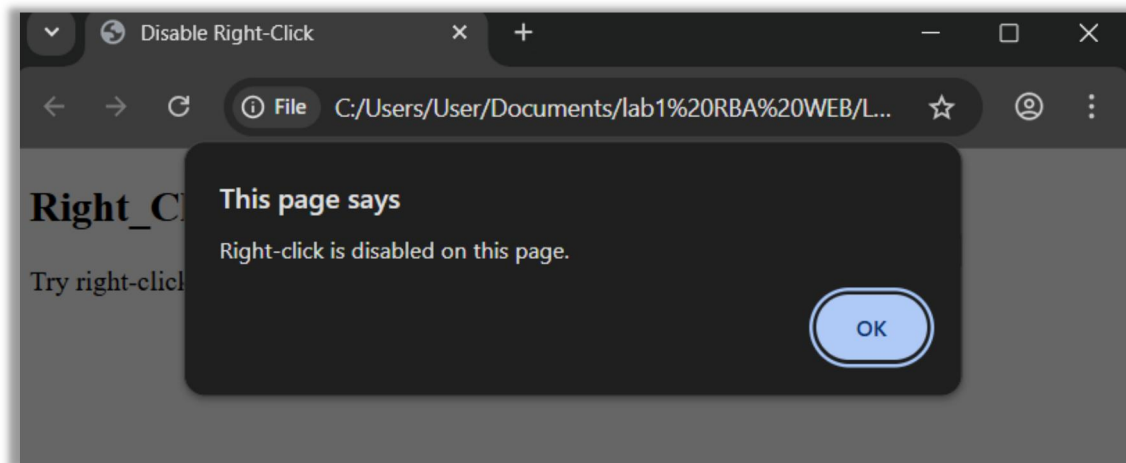
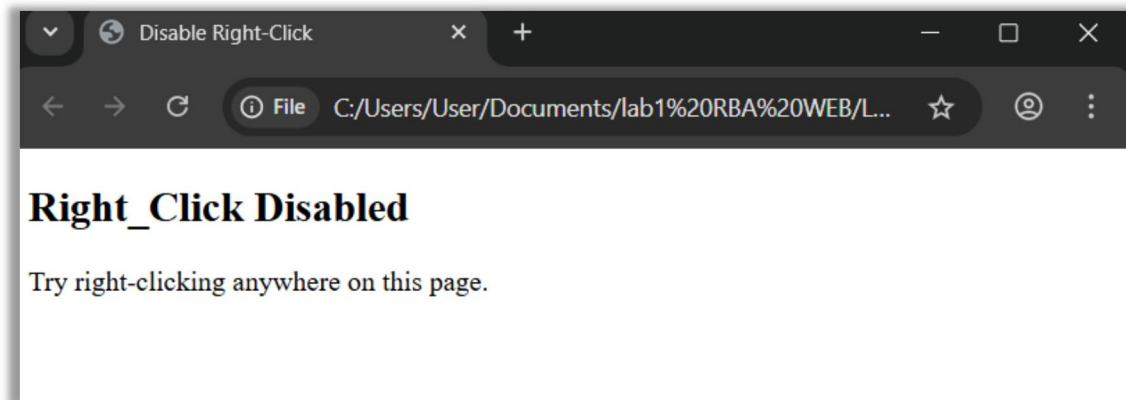
6.7.3 External References

Output:



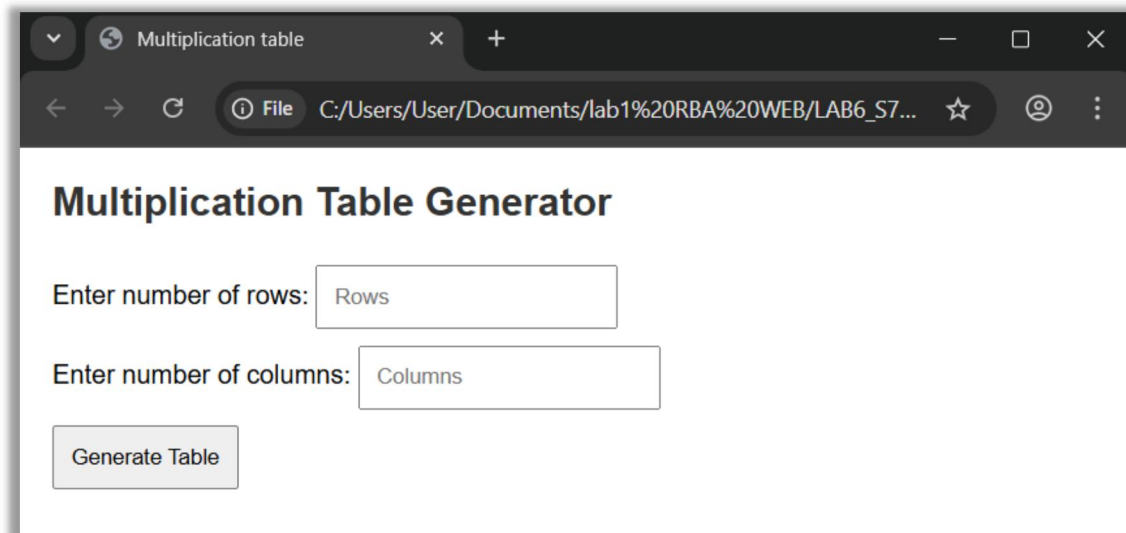
6.8 Working with events

Output:



6.9 Multiplication Table

Output:

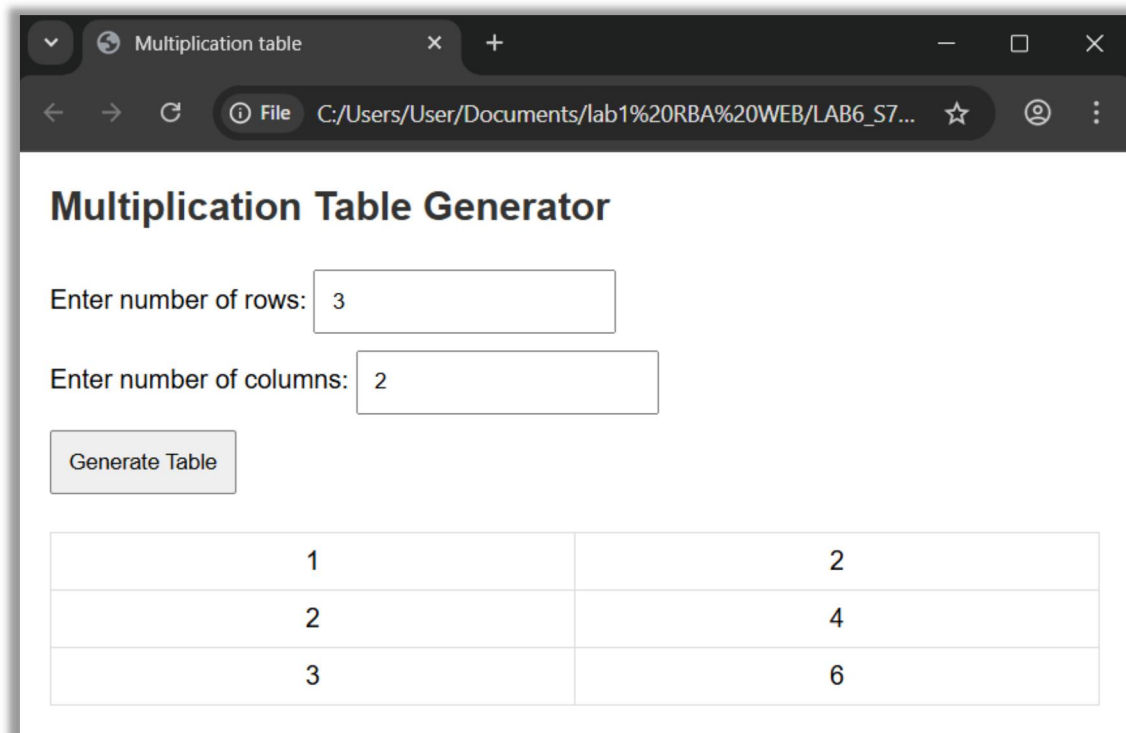


A screenshot of a web browser window titled "Multiplication table". The address bar shows a local file path. The page content includes a heading "Multiplication Table Generator", two input fields labeled "Enter number of rows:" and "Enter number of columns:", and a "Generate Table" button. The input fields contain the placeholder text "Rows" and "Columns" respectively.

Multiplication Table Generator

Enter number of rows:

Enter number of columns:



A screenshot of the same web browser window after the "Generate Table" button has been clicked. The input fields now contain the values "3" and "2". Below the inputs, a 3x2 multiplication table is displayed.

Multiplication Table Generator

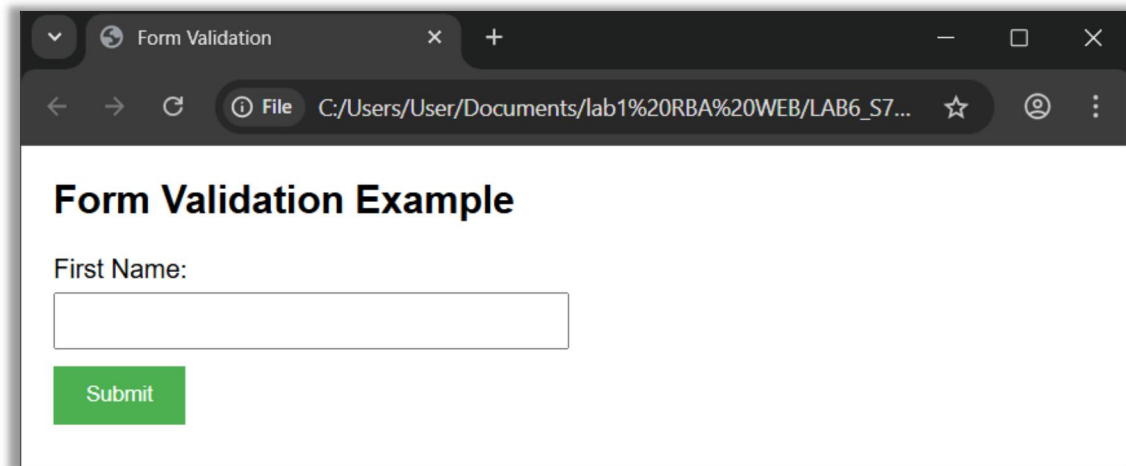
Enter number of rows:

Enter number of columns:

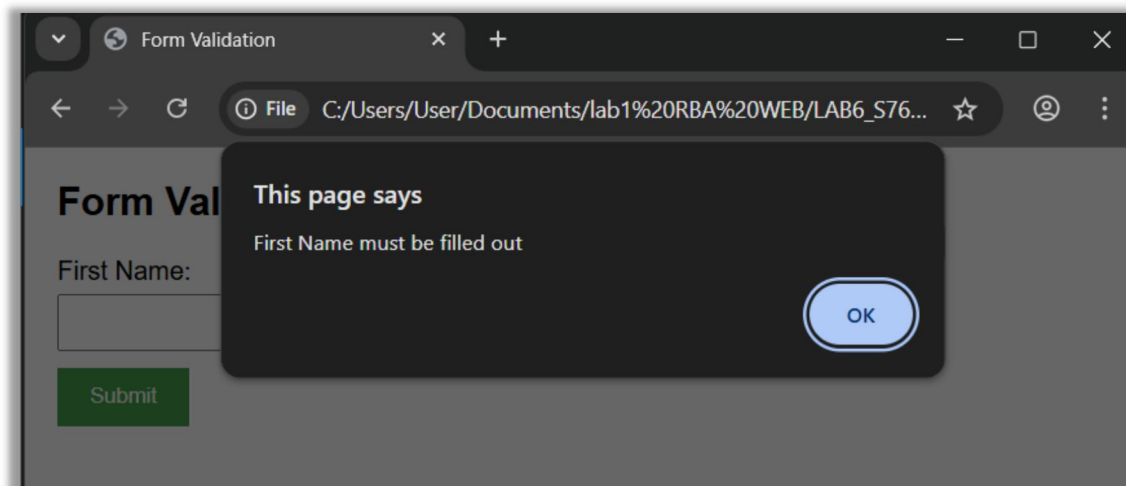
1	2
2	4
3	6

6.10.1 Form Validation

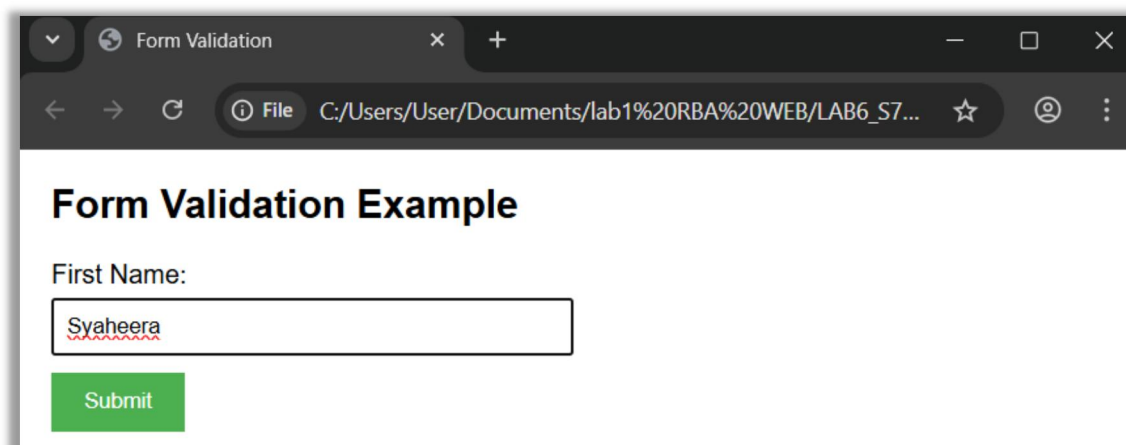
Output:



The screenshot shows a web browser window with the title 'Form Validation'. The address bar displays the file path 'C:/Users/User/Documents/lab1%20RBA%20WEB/LAB6_S7...'. The page content includes a heading 'Form Validation Example', a label 'First Name:', an empty text input field, and a green 'Submit' button.



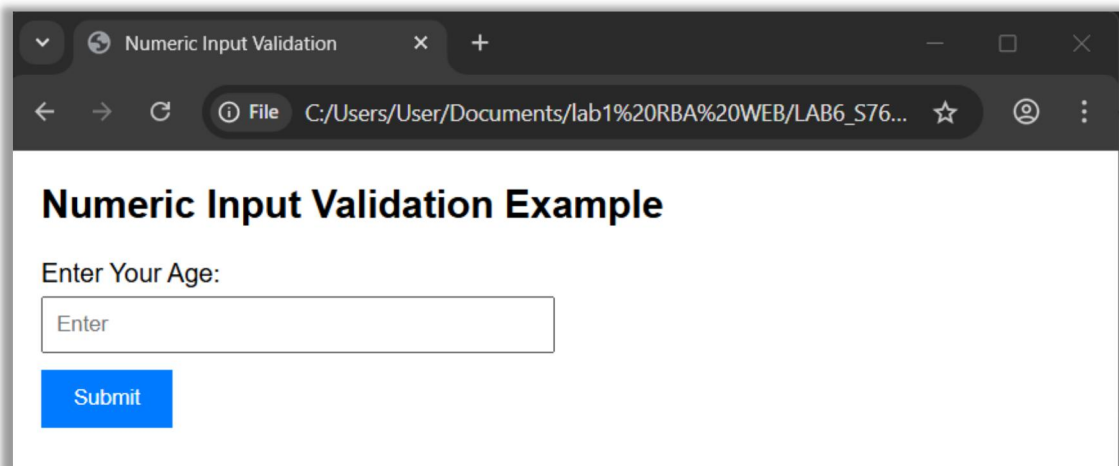
The screenshot shows the same web browser window, but with a validation error message displayed in a dark overlay. The message reads 'This page says' followed by 'First Name must be filled out'. There is an 'OK' button in the bottom right corner of the message box. The background form is dimmed.



The screenshot shows the web browser window with the form. The 'First Name' input field now contains the text 'Syaheera'. A red wavy underline is visible under the text. The green 'Submit' button remains below the input field.

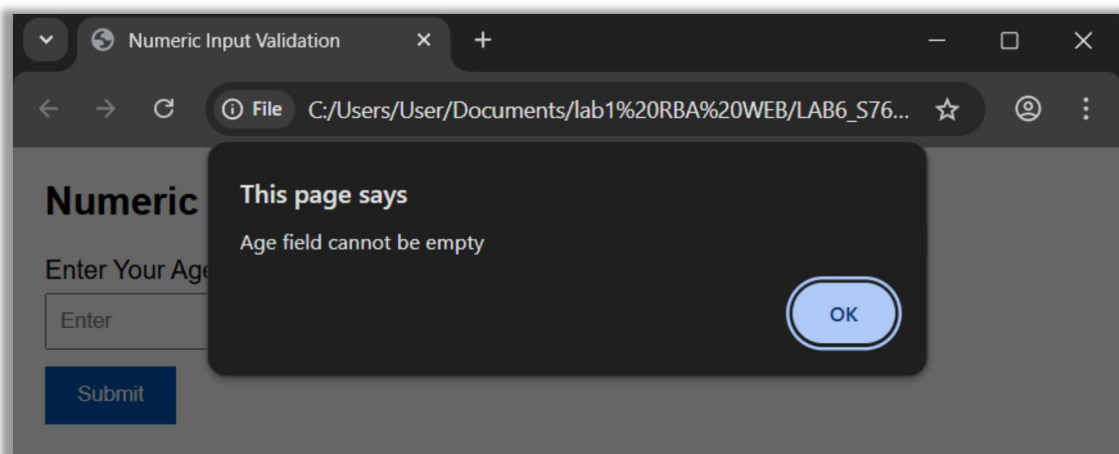
6.10.2 Numeric Input Validation

Output:



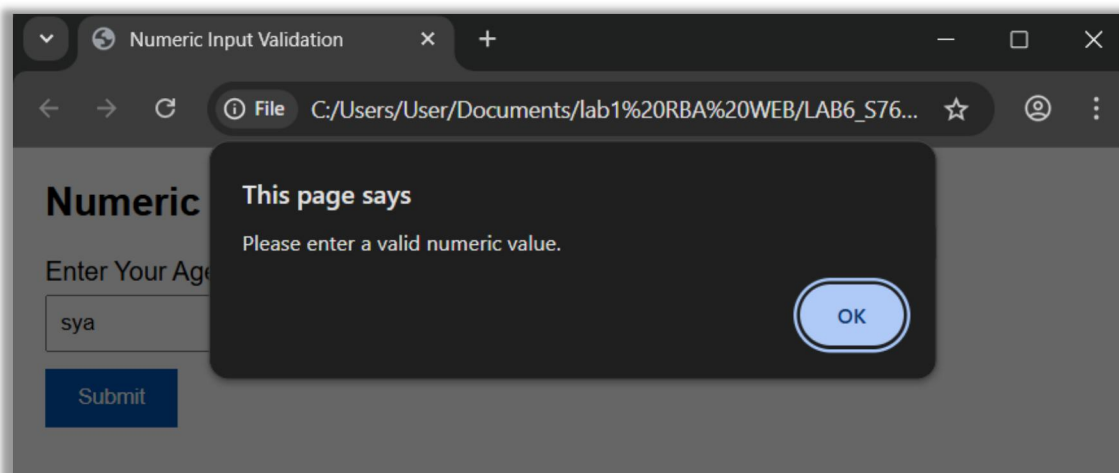
A screenshot of a web browser window titled "Numeric Input Validation". The address bar shows the file path "C:/Users/User/Documents/lab1%20RBA%20WEB/LAB6_S76...". The page content includes the heading "Numeric Input Validation Example", the label "Enter Your Age:", an input field containing the text "Enter", and a blue "Submit" button.

Leave the input field empty and click submit:



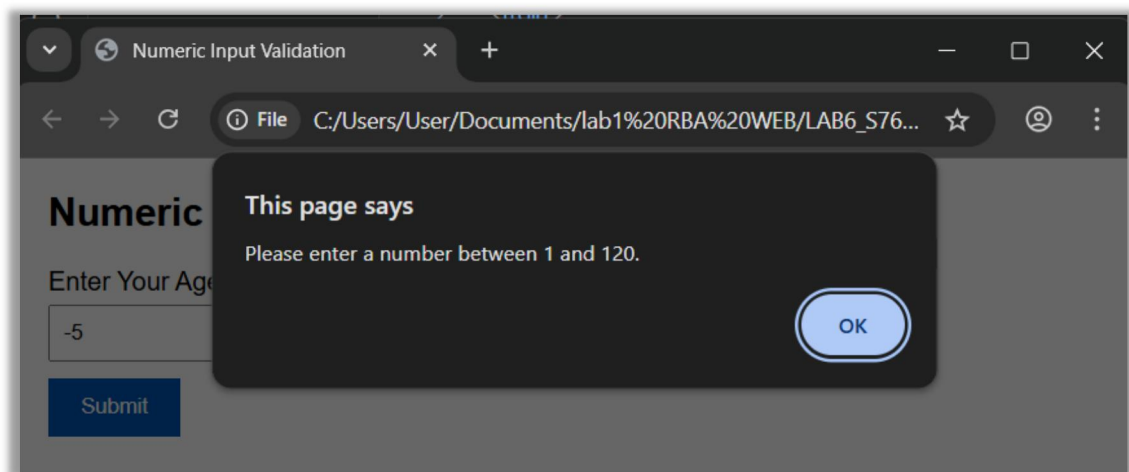
A screenshot of the same web browser window. The input field is now empty. A dark modal dialog box is displayed in the center of the screen with the title "This page says" and the message "Age field cannot be empty". An "OK" button is located at the bottom right of the dialog box. The "Submit" button on the page is now dark blue.

Enter a non-numeric characters and click submit:

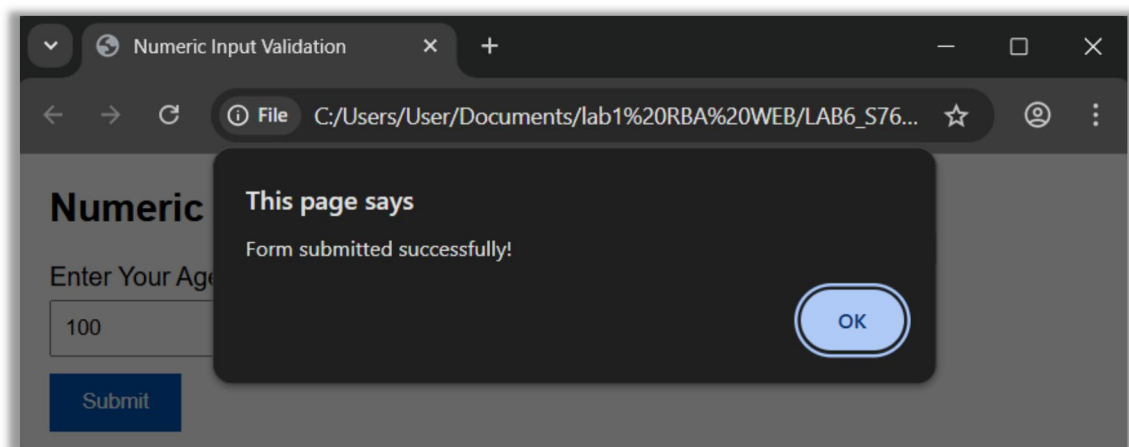


A screenshot of the same web browser window. The input field now contains the text "sya". A dark modal dialog box is displayed in the center of the screen with the title "This page says" and the message "Please enter a valid numeric value.". An "OK" button is located at the bottom right of the dialog box. The "Submit" button on the page is now dark blue.

Enter a number outside the range between -5 and 200:

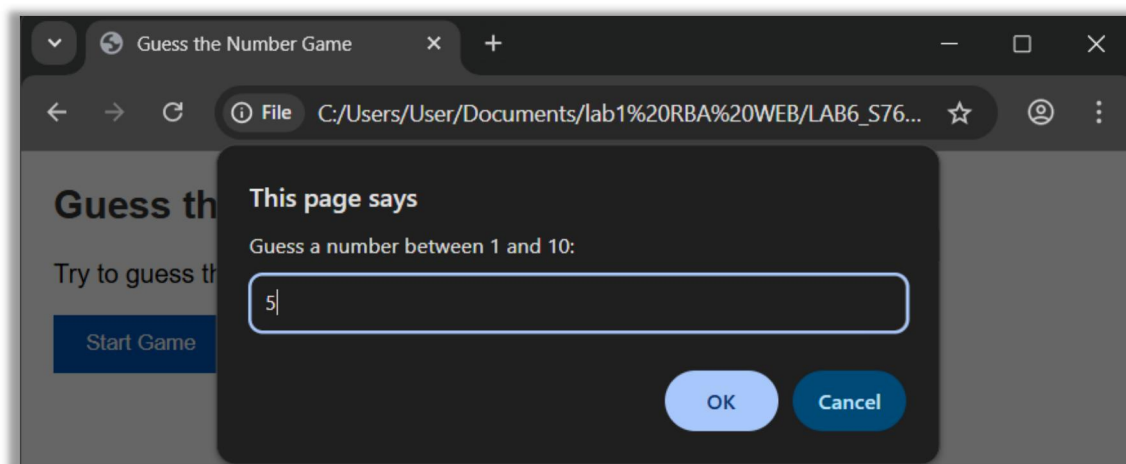
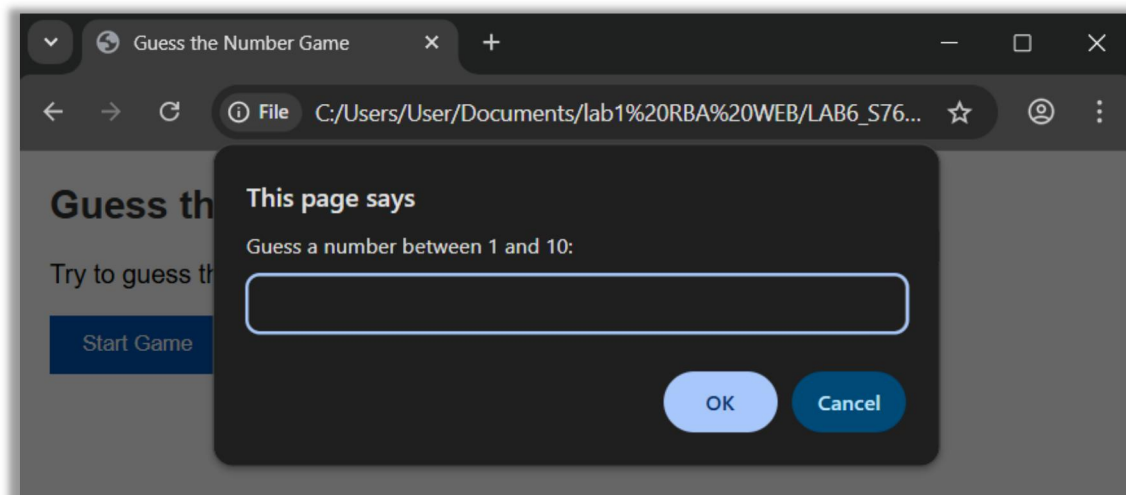
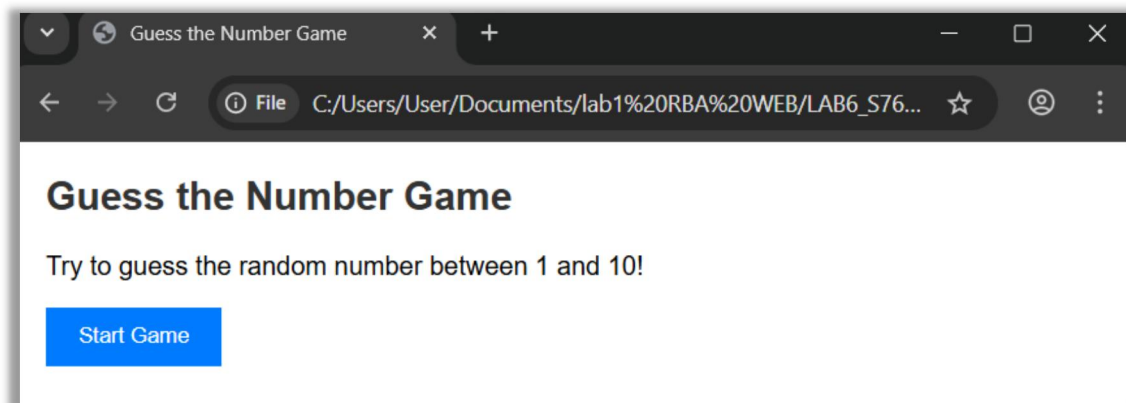


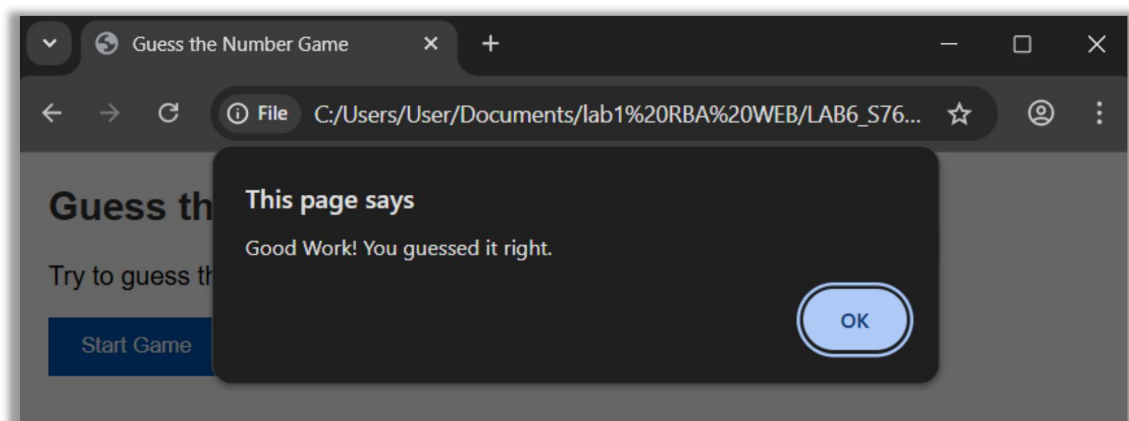
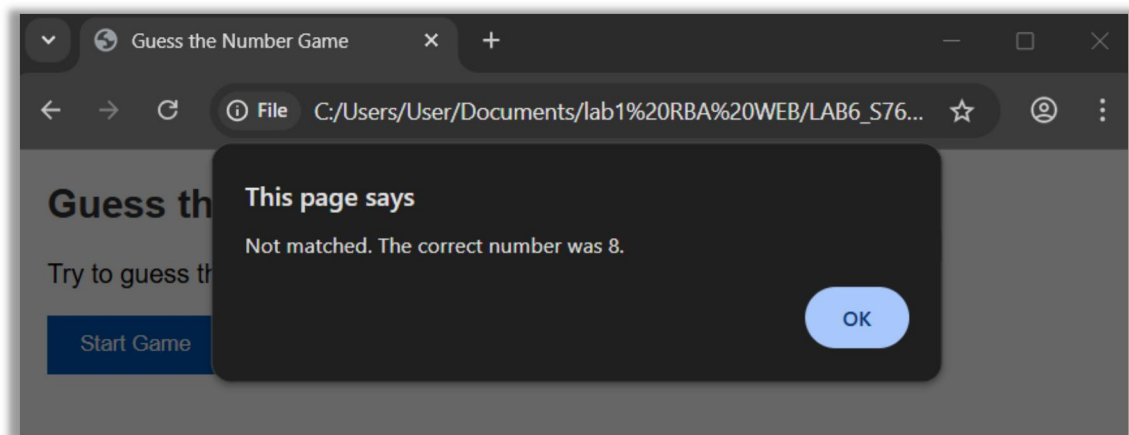
Enter a valid number and click submit:



6.11.1 Guess a Number

Output:





6.12 Lab Exercise 6

In this exercise, you will create a dynamic quiz application using HTML, CSS, and JavaScript. This task will help you integrate fundamental web development concepts to build an engaging user experience.

You will implement multiple-choice questions, a timer to challenge users, and real-time feedback for correct and incorrect answers. Additionally, questions will be randomized to enhance interactivity. By completing this lab, you will strengthen your skills in structuring content, styling with CSS, and adding interactivity with JavaScript

6.12.1 Quiz Application

Instruction

1. Create the following files in your project directory:

- index.html (for the HTML structure)
- style.css (for the CSS styling)
- script.js (for the JavaScript functionality)

2. In your index.html, structure the page with the following elements

- A header that introduces the quiz.
- A section to display the questions, answer options, and a timer.
- A section to show feedback (correct/incorrect) after each answer.
- A button to submit answers and navigate between questions.
- A footer to display the user's score after finishing the quiz.

3. In your style.css file, style the quiz application to make it visually appealing

4. In your script.js, implement the following javascript functionalities

- Questions Array: Stores the questions, possible options, and the correct answer.
- shuffle Questions(): Randomly shuffles the questions to ensure each quiz is different.
- startTimer(): Sets a countdown timer for answering each question.
- displayQuestion(): Displays the current question and answer options.
- checkAnswer(): Checks whether the selected answer is correct and updates the score.
- next Question(): Moves to the next question, resetting the timer and updating the score.

- `startQuiz()`: Starts the quiz by shuffling the questions, displaying the first question, and starting the timer.

5. Save index.html, style.css and script.js and zip the file

6. Submit the files via elearn platform.

Output:

